Week #: 12 (16/11/2026 - 20/11/2026)

Subject: Pre-Calculus Math **Unit Title: Radical Equations and Polynomial Functions** **Grade: 11**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1: Solving Radical Equations: Square Roots	• I can isolate a square-root expression and square both sides. • I can solve simple radical equations and check solutions in the original equation.	• Graph interpretation and guided sketching from degree, leading coefficient and zeros. • Partner comparison of cubic and quartic end behaviour and turning points. • Support: graph templates; extension: repeated zeros and multiplicity.	• Lesson 2.2 Polynomial Functions PowerPoint • Practice Sheet 2.2 on Microsoft Teams • Desmos / graphing calculator and graph paper
Lesson 2: Solving Radical Equations: Extension	• I can solve multi-step radical equations, including equations requiring repeated squaring. • I can identify and reject extraneous solutions by substitution.		
Lesson 3: Sketching Cubic Functions	• I can identify degree, leading coefficient, end behaviour, zeros and intercepts of cubic functions. • I can state the maximum of 3 real zeros and 2 turning points, then sketch a cubic graph.		
Lesson 4: Sketching Quartic Functions	• I can determine quartic end behaviour and identify zeros, repeated zeros and multiplicity. • I can state the maximum of 4 real zeros and 3 turning points and sketch a quartic graph.		
Homework	Complete Practice Sheet 2.1 and Practice Sheet 2.2, available on Microsoft Teams.		
Teacher's notes/ Reminders	• Check Microsoft Teams for Practice Sheets 2.1 and 2.2. • Bring a calculator, charged device and graphing materials. • Emphasise checking radical solutions and distinguishing maximum from actual turning points.		

Technology Integration	Desmos, Microsoft Teams, graphing calculator, GeoGebra and coordinate grids.
Cross-curricular links	Physics: Radical equations in distance and measurement models. Engineering: Cubic and quartic functions in design and modelling.
External Assessment Integration	SAT/ACT/AP Precalculus-style radical equation solving, extraneous solutions, polynomial end behaviour, zeros and turning points.
Wellbeing/ ATL Skills	Critical Thinking: Checking for extraneous solutions and justifying polynomial graph features. Communication: Explaining graph sketches. Self-Management: Completing practice sheets.
Standards	CCSS.HSA-REI.A.2 Solve simple radical equations and identify extraneous solutions. CCSS.HSF-IF.C.7c Graph polynomial functions and identify zeros. CCSS.HSF-IF.B.4 Interpret key features of graphs.
Inquiry/ TOK Question	How can algebraic solutions and the leading term help us predict the shape of a function?
Success Criteria (Measurable)	**Lesson 1: Solving Square-Root Equations** • All: isolate a radical. • Most: solve by squaring and check answers. • Some: explain extraneous solutions. **Lesson 2: Multi-Step Radical Equations** • All: follow a structured solution process. • Most: solve and verify multi-step equations. • Some: solve equations requiring repeated squaring. **Lesson 3: Cubic Functions** • All: identify degree and leading coefficient. • Most: describe end behaviour and sketch using zeros. • Some: explain why a cubic has at most 2 turning points. **Lesson 4: Quartic Functions** • All: recognise quartic end behaviour. • Most: sketch using zeros and turning points. • Some: explain repeated zeros and multiplicity.
National Identity Domain	Conservation الحفاظ علي البيئة Global Understanding التفاهم العالمي Heritage التراث

	Area: UAE Innovation and Future Readiness Area: UAE culture and heritage Area: UAE Economic Development **Area:** Responsible Citizenship and Innovation
	Implementation: • Students explore polynomial models used in UAE infrastructure and engineering. • Students connect careful checking of radical equations to accurate scientific calculations. • Graphing activities support UAE innovation and future-readiness.